S1.1 Introduction to the particulate nature of matter

The big picture

Guiding question(s)

How can we model the particulate nature of matter?

Keep the guiding question in mind as you learn the science in this subtopic. You will be ready to answer it at the end of this subtopic. The guiding questions require you to pull together your knowledge and skills from different sections, to see the bigger picture and to build your conceptual understanding.

Models are particularly useful in the study of chemistry as they allow us to visualise on a larger scale (the macroscopic scale) that which is usually too small to be seen with the unaided eye (the microscopic scale). However, the use of models in science can also lead to misconceptions. Take the planetary model shown below, for example (**Figure 1**). At first, it could be argued that it is a simple model to help students remember the order of the planets in the solar system. But a closer look at the model may reveal some inconsistencies – can you spot what they are?

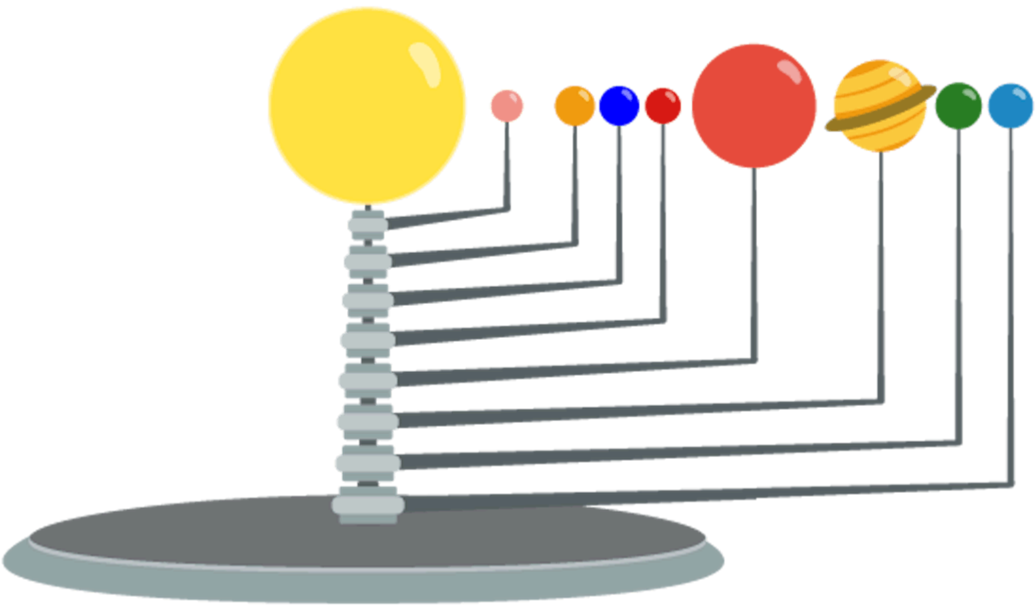

Figure 1. A model of the planets in the solar system — can you spot the inconsistencies in the model?

More information for figure 1

The image displays a drawing of a mechanical solar system model. It features a yellow sphere representing the Sun, mounted on a stand. Radiating outward from the Sun are nine smaller spheres symbolizing the planets of the solar system. Each planet is connected to the Sun via a thin line. The planets vary in size and color, indicating differences in perceived sizes and characteristics. The arrangement of the planets shows how they are orbiting the Sun, similar to an orrery model. While this model visually represents the solar system, the spacing and scale are not to proportion, which might lead to misconceptions if used for educational purposes.

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> The obvious inconsistencies with this model are that the distances between the planets are not correct and that the size of the planets are not to the correct scale.

There are other questions that you could ask about the model:

- How useful would you say this model is?
- What improvements could be made?
- Do you think all scientific models have similar issues?

In the next section, you will learn about the particle model of solids, liquids and gases. In the particle model, round spheres are used to represent the smallest units of matter, particles. Being able to visualise the particles makes it easier to understand changes of state such as melting or boiling. But there are questions that could be raised about this model:

- Are all particles really spherical in nature?
- Can one model correctly represent all types of matter?

These are the types of questions that you should be asking when a model is used in the DP chemistry course. You could also think of possible improvements to the model in question.

Concept

Structure is one the two main concepts in the DP chemistry course, with the other being reactivity.

Structure in chemistry is concerned with the way in which the particles of a substance are arranged. In this course, you will learn the importance of how the structure of different substances can affect their physical properties, such as its melting and boiling points, as well as the chemical reactivity.

The second concept, **reactivity**, is concerned with how and why substances interact with each other. This concept will be explored later in the course.

Prior learning

Because this is the first subtopic in the textbook, there is no required prior learning as such. The following sections deal with states of matter and changes of state so you might have some experience of these from your everyday life such as boiling water when cooking or the melting of butter. It may be helpful to keep these in mind as you read the following sections.

Practical skills

Once you have completed this subtopic, you can apply your knowledge of mixtures by going to Practical 1a: Separation techniques (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/separation-techniques-1a-id-43173/) and Practical 1b: Purification technique: recrystallisation (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/purification-technique-recrystallisation-1b-id-46713/).